

Lab 3: Architecture Selection & Component Modelling

Prahas Bodanapati – PES1UG24CS327 – Lab3 – 11 September 2026

System: Hyperlocal Courier Dispatch & Tracking Engine — Problem Statement #23

Architecture Selection

We chose **Microservices Architecture** for the Hyperlocal Courier Dispatch & Tracking Engine.

Architectural Choice

The system is divided into independently deployable services such as Order Manager, Payment Service, Rider Matching Service, Live Tracking Service, and OTP/Delivery Verification, with client applications and external gateways/providers around them.

Two Scenario-Related Reasons

1. Real-time scalability: Rider matching and live tracking can be scaled independently because the requirements demand nearest-rider matching within a short dispatch window and live GPS telemetry with under 2-second latency. **2. Fault isolation and availability:** The dispatch/matching service has a 99.5% uptime target. Separating services reduces the chance that a failure in payment, OTP verification, or another service takes down the entire platform and supports independent deployment and recovery.

Security Advantage

Service boundaries allow sensitive responsibilities to be isolated. The Payment Service can communicate with the Payment Gateway without exposing payment details to unrelated services, while OTP/Delivery Verification validates the recipient before the order becomes Delivered. API Gateway controls and authenticates client-facing requests.

Performance Benefit

Microservices allow high-load functions to scale separately. Rider Matching and Live Tracking can receive additional instances during peak demand, while lightweight API and order operations remain independently scaled. This supports fast dispatch and the required low-latency telemetry.

Component & Interface Summary

The component diagram shows the given Order Manager and Payment Service plus API Gateway, Rider Matching Service, Live Tracking Service, OTP/Delivery Verification, Courier Database, and client/external components. Key interfaces cover order requests, payment authorization, rider matching/dispatch, live telemetry, OTP verification, database access, and map/location services.